

rb_vi_bench

Reduced-basis dual cones for variational inequalities:
algorithms, metrics, and measured results

Benchmark report

September 7, 2026

Abstract

This report documents the benchmark assembled from three source repositories for reduced-basis (RB) model order reduction of contact problems. It states the problem setting, every selection algorithm under test, the exact computational definition of every metric, and the results measured on the three retained contact datasets. The object under study throughout is the *dual cone*: the reduced set from which Lagrange multipliers are drawn, on which non-negativity must hold. Sources: [BEE20] = Benaceur, Ern, Ehrlacher (ha1-02081485v2); [NDEE22] = Niakh, Drouet, Ehrlacher, Ern (ESAIM M2AN, 2022). Both papers number their algorithms and equations independently, so every citation below carries its paper tag.

1 Problem setting

The high-fidelity (HF) problem is a discrete variational inequality of contact type: find the displacement $u(\mu) \in \mathbb{R}^N$ and multiplier $\lambda(\mu) \in \mathbb{R}^m$ solving the saddle-point system

$$A u(\mu) + B(\mu)^\top \lambda(\mu) = f(\mu), \quad B(\mu) u(\mu) \leq g(\mu), \quad \lambda(\mu) \geq 0, \quad (1)$$

with complementarity between the last two. The sign condition $\lambda \geq 0$ is physical: contact pressure pushes, it does not pull.

Reduction replaces $\mathbb{R}^m$ by a low-dimensional set for λ . That set cannot be a *subspace*: a subspace closed under negation cannot enforce $\lambda \geq 0$. It must be a finitely generated **convex cone**

$$W_R^+ = \text{span}_+ \{\xi_1, \dots, \xi_R\} = \left\{ \sum_{r=1}^R c_r \xi_r : c_r \geq 0 \right\}, \quad \xi_r \geq 0. \quad (2)$$

With $\xi_r \geq 0$ and $c_r \geq 0$, membership of the cone *implies* the sign condition. This is the single structural reason the whole family of methods exists, and it is why POD is inadmissible here: its modes have mixed signs ([BEE20] §5).

Two derived objects are used repeatedly. The **full snapshot cone** $K_{\text{full}} = \text{span}_+ \{\theta_1, \dots, \theta_Q\}$ is everything the training multipliers generate; and the **non-negative orthant** $W^+ = \{x \in \mathbb{R}^m : x \geq 0\}$ is the largest admissible set of all. Every method produces some W_R^+ with $W_R^+ \subseteq W^+$, but—importantly—*not* necessarily $W_R^+ \subseteq K_{\text{full}}$.

The **cone projection** $\Pi_K(x) = \arg \min_{y \in K} \|x - y\|$ is computed by non-negative least squares (NNLS) on the generator matrix. It is non-linear in x , which is what makes every error below more expensive than a linear projection and is the unit in which offline cost is counted.

2 Algorithms under test

All greedy methods are *hierarchical*: the cone at cardinality R is a sub-cone of the cone at $R+1$. This is verified rather than assumed (§4). NMF is not hierarchical, and that difference is visible in several metrics.

2.1 CPG — Cone-Projected Greedy ([BEE20], Algorithm 2)

Start from $K_0 = \{0\}$. At step r , select the training snapshot worst represented by the cone built so far, and append it unchanged:

$$q_r \in \arg \max_q \|\theta_q - \Pi_{K_{r-1}}(\theta_q)\|, \quad K_r = \text{span}_+\{\theta_{q_1}, \dots, \theta_{q_r}\}. \quad (3)$$

Because $K_0 = \{0\}$ gives $\Pi_{K_0} \equiv 0$, the first step reduces to $q_1 \in \arg \max_q \|\theta_q\|$, which is [BEE20] Eq. (56). The generators are *selected snapshots*, so $K_R \subseteq K_{\text{full}}$ holds by construction.

2.2 mCPG — modified CPG ([NDEE22], Algorithm 2)

Same selection loop, but the appended generator is a *residual* rather than the snapshot: $\nu_r = (\theta_q - \Upsilon_r) / \|\cdot\|$ with $\Upsilon_r \in K_{r-1} \cap (\theta_q - W^+)$. The second constraint forces $\nu_r \geq 0$, i.e. membership of W^+ , which is the property the method needs. It does *not* force membership of K_{full} : a difference of two elements of $\text{span}_+\{\theta\}$ need not lie in $\text{span}_+\{\theta\}$. This is measured directly (§5).

2.3 ADG — Batch Normalized Angular-Defect Greedy

Not in either paper; contributed by the `greedy_algos` source. Two departures from CPG:

- **Angular selection.** It ranks candidates by the angle between a snapshot and its cone projection, $\theta_K(x) = \angle(x, \Pi_K(x))$, rather than by residual magnitude. On the normalized set $S_{\text{norm}} = \{x/\|x\|\}$ the two induce the same ordering, since the residual is $\sin \theta_K(x)$; un-normalized they differ. ADG is therefore *defined* on S_{norm} , and the tolerance becomes a genuine per-snapshot relative bound $\varepsilon \in (0, 1)$ rather than one shared absolute threshold $\varepsilon \max_q \|\theta_q\|$.
- **Batch admission.** Every candidate attaining the maximal defect θ_{max} enters in one round, not just one. This is what the method’s Theorem 3.5 certificate rests on.

Three ADG variants are benchmarked, each differing from the base in exactly one respect, so that any measured gap is attributable:

key	varies	against
<code>adg</code>	— (reference form)	
<code>adg_momentum</code>	stopping rule: $ e(p) - e(p-1) /e(p-1) \leq \varepsilon$	<code>adg</code>
<code>adg_k0</code>	initialization: $K_0 = \{0\}$, i.e. CPG’s first step	<code>adg</code>

The base ADG initializes not from $K_0 = \{0\}$ but from the *pair* of snapshots at the largest mutual angle. This is its one departure from both papers, and `adg_k0` is its ablation: the first generator is instead $\arg \max_q \|\theta_q\|$ taken on the *unnormalized* snapshots ([BEE20] Eq. (56)), normalized, after which the identical angular loop runs. Note that $K_0 = \{0\}$ does not by itself define ADG’s first step: with the zero cone every defect is a right angle, all candidates tie, and batch admission would take the entire training set at once — Eq. (56) is the tie-break the papers supply. A consequence of the pair start is that base ADG has *no* $R = 1$ state; its trajectory is $R = 2, 3, 4, \dots$

2.4 Baselines

- **NMF** ([BEE20] §6.4): optimized non-negative atoms, free to sit anywhere in W^+ . A genuine competitor, but cardinality-only — [BEE20] §5’s objection is precisely that “the user does not specify an error tolerance but only the cardinality” — and non-deterministic in its initialization.

- **Orthant** (reference, not a competitor): canonical directions e_i selected along mCPG’s own iteration, so the only difference from mCPG is the generator itself. Its pairwise angles are exactly 90° , the widest any cone in W^+ can be. Its reported cost is the mCPG ordering run, not the (free) cone assembly.
- **POD** (*negative* control): the best unconstrained reconstruction at a given cardinality, and a direct measurement of the sign violation that motivates the cone construction. Its non-negativity check is *expected* to fail.

3 Metrics: how each is computed

3.1 Precision

For a cone W_R^+ with generator matrix Ξ , each snapshot is projected by NNLS and the residual is reported relative to a normalizer. Two normalizers are reported because they answer different questions:

$$\underbrace{\frac{\|\theta_q - \Pi(\theta_q)\|}{\max_{q'} \|\theta_{q'}\|}}_{\text{shared denominator}} \quad \text{and} \quad \underbrace{\frac{\|\theta_q - \Pi(\theta_q)\|}{\|\theta_q\|}}_{\text{per-snapshot}}. \quad (4)$$

The shared denominator keeps the number comparable against the tolerance that produced it; the per-snapshot form is what ADG’s tolerance actually bounds. They diverge whenever snapshot magnitudes spread. Train and test are reported separately.

Non-negativity is checked on the reconstruction: $\text{nn_max_violation} = \max(0, -\min_i \hat{\lambda}_i)$.

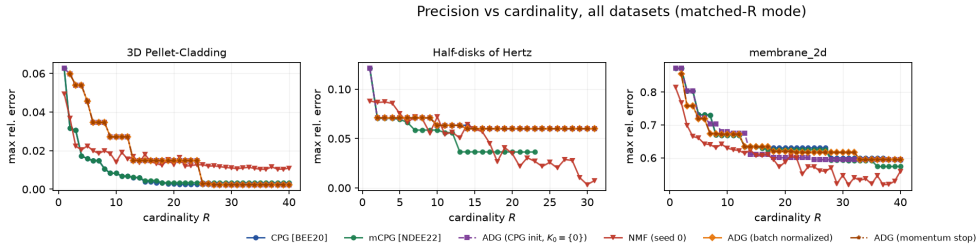


Figure 1: Test projection error against cardinality, all three datasets. Compressibility differs by orders of magnitude: Half-disks and Pellet-Cladding fall steeply, membrane barely moves — 125 snapshots on 497 dual dofs with a moving contact patch is simply not compressible to $R \leq 40$, which is a property of the problem and not of any method.

3.2 Performance (offline cost)

Wall-clock is machine-dependent, so the primary cost is a *count of constrained solver calls* — NNLS, `lsq_linear` and `minimize` — captured by instrumenting the solvers during a fit. One caveat governs its reading: it counts *constrained* solves only, so NMF records exactly 0, meaning “issues none”, not “is free”. ADG likewise records 0 at $R = 1, 2$, because it seeds from a Gram-matrix argmin and NNLS first appears at $R = 3$.

3.3 Stability

- **Gram conditioning:** $\kappa(\hat{\Xi}^\top \hat{\Xi})$ on *column-normalized* generators. Normalizing is essential: without it the number is dominated by snapshot magnitude spread, and [BEE20] (raw generators) and [NDEE22] (normalized) would be incomparable. Undefined for $R < 2$ and reported as such.

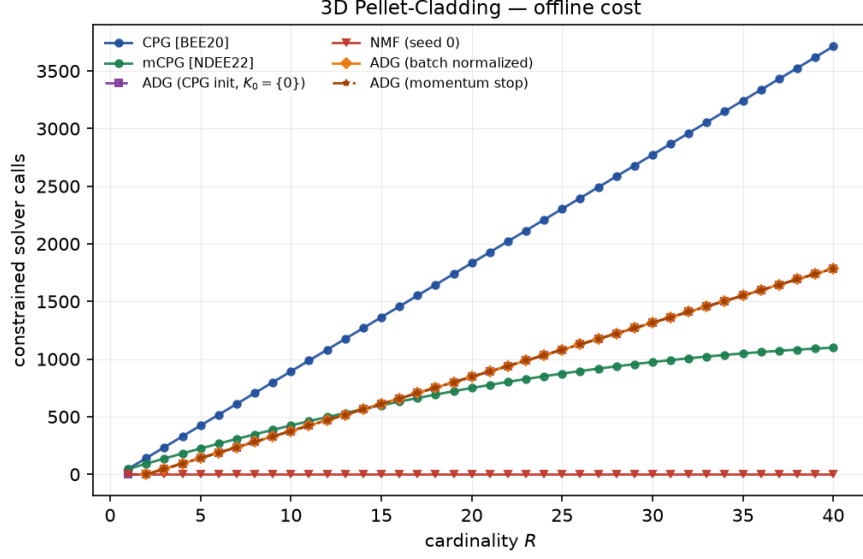


Figure 2: Offline cost on 3D Pellet-Cladding. CPG grows about twice as fast as ADG in constrained solves; mCPG plateaus once its residual stops finding new directions; NMF sits on the zero line throughout, which means it issues no *constrained* solves, not that it is free.

- **Orthogonality** e_{orth} ([NDEE22] Eq. 41): $\|(I - \Pi_{K_r})(\nu_{r+1})\|$, the sine of the angular defect of each accepted generator.
- **Determinism**: across-seed spread, which is the non-determinism [BEE20] §5 raises against NMF.

3.4 Cone geometry

This family scores the cone against K_{full} rather than against the finite snapshot set, by Monte-Carlo sampling of Dirichlet(1) conic combinations.

$$\text{missed (too small)} : \mathbb{E}_{x \sim K_{\text{full}}} \|x - \Pi_{W_R^+}(x)\|, \quad (5)$$

$$\text{excess (too large)} : \mathbb{E}_{x \sim W_R^+} \|x - \Pi_{K_{\text{full}}}(x)\|, \quad (6)$$

$$\text{two-sided} : \frac{1}{2}(\text{missed} + \text{excess}). \quad (7)$$

Neither direction implies the other: a cone can cover K_{full} perfectly while extending far beyond it, or sit strictly inside while missing most of it. The two-sided figure is the *average*, not the maximum. A maximum reports only whichever direction is larger and so silently changes which quantity it plots from one method to the next; the average is the same functional for every method, keeps both faults in view, and is still a metric on cones (symmetric, zero only when both vanish, triangle inequality inherited).

Extent (how much space the cone encloses)

Cut W_R^+ by a hyperplane $\langle x, u \rangle = h$ shared by every method, with $u = (1, \dots, 1)/\sqrt{m}$ — strictly positive on $W^+ \setminus \{0\}$, so the cut is transversal for every admissible cone, and no coordinate is privileged. The section is the convex hull of $p_i = h \xi_i / \langle \xi_i, u \rangle$, and what is reported is its **mean width**, normalized by that of K_{full} 's section:

$$\text{extent}(W_R^+) = \frac{w(S_R)}{w(S_{\text{full}})}, \quad w(S) = \mathbb{E}_{g \sim \mathcal{N}(0, I_m)} \left[\max_i \langle p_i, g \rangle - \min_i \langle p_i, g \rangle \right]. \quad (8)$$

Three properties make this the right statistic, and each rules out an alternative that was tried and discarded:

- *Measured in the dimension of the space, not the number of generators.* The ξ_i are R rays in $\mathbb{R}^m$ with $R \ll m$; they are not a basis. Taking the section's $(R-1)$ -volume builds the method's cardinality into the yardstick.
- *Monotone in R by construction.* $W_R^+ \subseteq W_{R+1}^+$, so the section can only grow; more vertices means a maximum over a superset, so the integrand rises pointwise in g . The Gaussian directions are drawn once from a fixed seed, so monotonicity holds per realization and Monte-Carlo noise cannot manufacture a decrease.
- *Positive on a thin cone*, where every ambient volume, solid angle or covered-sphere fraction is identically zero.

Values above 1 are meaningful: the cone is wider than K_{full} , which requires leaving it. Scale-invariance ($\text{span}_+\{c\xi\} = \text{span}_+\{\xi\}$) means generator normalization cannot affect it.

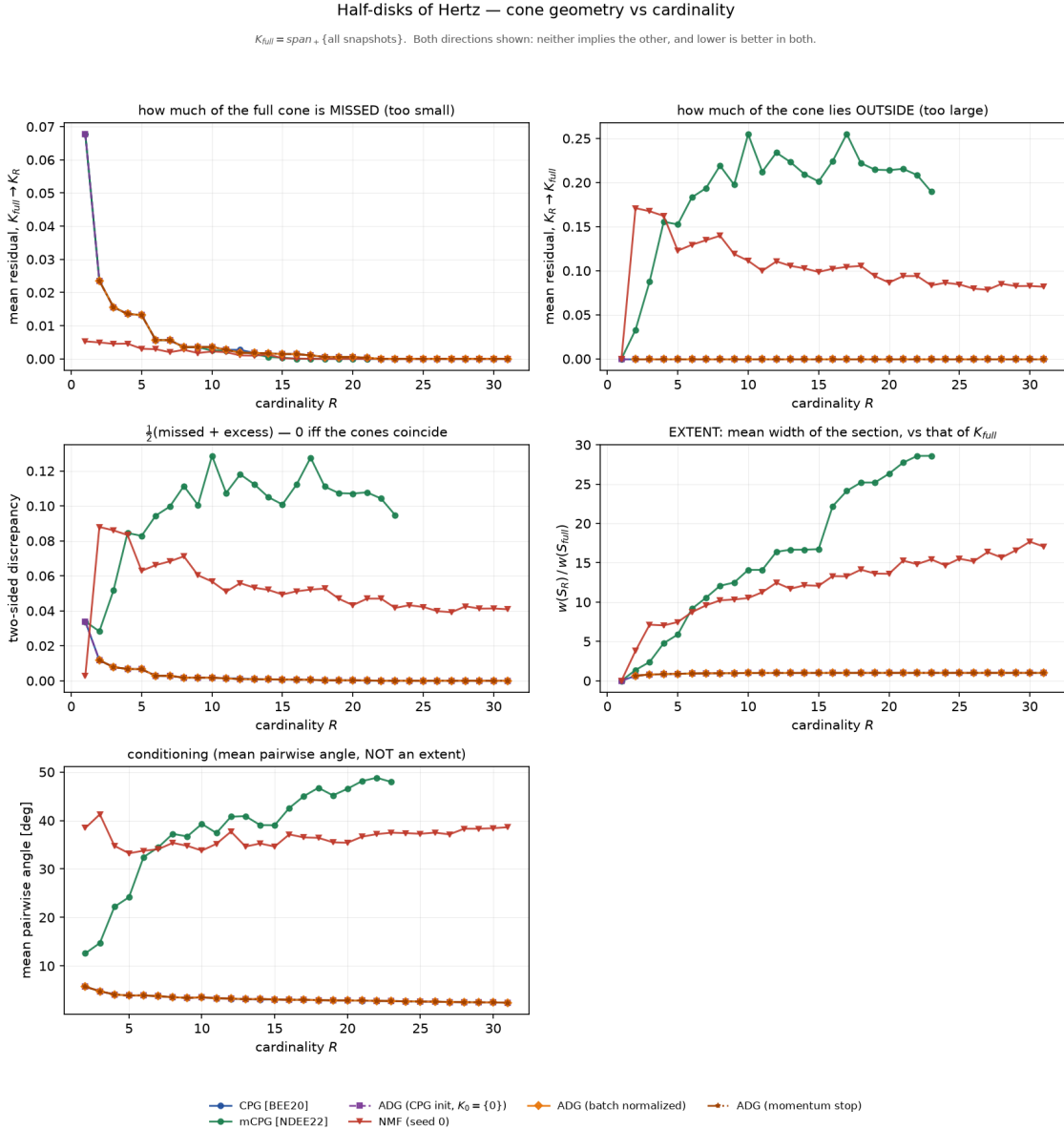


Figure 3: The five cone-geometry panels on Half-disks of Hertz. Read the two residual panels together: CPG and ADG sit flat at zero in *excess* because a cone spanned by snapshots cannot leave K_{full} , while mCPG lifts to 0.26. Extent and aperture disagree by design — aperture has mCPG rising smoothly while extent shows what the enclosed region actually does.

Aperture (conditioning, *not* extent)

Mean pairwise angle between normalized generators. Retained as a conditioning diagnostic only. It is not a measure of enclosed space: it saturates once generators are mutually well separated, and a generator added strictly *inside* the cone enlarges the region not at all while still moving the mean.

3.5 Solved error (metrics.online)

Every metric above scores a cone against snapshots; none solves anything. This one assembles and solves the reduced saddle-point problem ([BEE20] Alg. 1 line 4) at each held-out parameter,

$$\hat{A} = V^\top A V, \quad \hat{B} = \Xi^\top B(\mu) V, \quad \min_{\alpha \geq 0} \frac{1}{2} \alpha^\top \hat{B} \hat{A}^{-1} \hat{B}^\top \alpha - \alpha^\top (\hat{B} \hat{A}^{-1} \hat{f} - \hat{g}), \quad (9)$$

and reports the relative error in *both* components against the HF solution. This is the *dual* form of the reduced problem, obtained by eliminating the primal coefficients in closed form; it is derived step by step in Appendix A, which also explains in what sense it is still a projection. The bound $\alpha \geq 0$ is the discrete form of span_+ , and it is why the recovered $\hat{\lambda} = \Xi \alpha$ is non-negative exactly rather than approximately. Projection error and solved error are different functionals of the same cone: the solve minimizes an energy over W_R^+ , it does not project the true multiplier onto it.

3.6 Cross-implementation agreement

Two independent transcriptions of CPG and of mCPG are retained deliberately. Agreement between them is checked at matched tolerance, so that a disagreement is a *finding* about an under-specified step rather than a rounding difference.

4 Experimental protocol

Datasets. Three contact problems on real geometry:

key	reported name	dim $\times$ n	geometry
fem_lambda	Half-disks of Hertz	57×50	mirrored contact line
physics	3D Pellet-Cladding	7676×94	76×101 surface grid
membrane_2d	membrane (2-D)	497×125	scattered 2-D patch

Two modes, never mixed. *Matched tolerance* gives every method the same ε and reports the R it needed; *matched cardinality* gives every method the same R . Only the second is used for comparison figures, because tolerances are not commensurable across methods (ADG’s is per-snapshot relative; CPG/mCPG’s is one shared absolute threshold).

One greedy trajectory. The cardinality sweep re-fits at each R . That is only legitimate if a greedy stopped at R yields exactly the first R generators of a longer run, so the property is asserted by test rather than assumed — and testing it found a real violation, since the upstream fixed-component helper answered $R = 1$ for ADG through a different selection rule.

Skips are recorded, never dropped. A cell that does not run emits a `skip_reason`, because “not measured here” and “scored badly here” must never be confusable.

Axes. All figure axes are linear, with one documented exception: the Gram condition number, which is ≥ 1 by definition (so the zero-dropping failure of a log axis cannot occur) and spans eleven decades before the basis goes singular.

5 Results

All numbers below come from the current run (817 of 840 cells; 1848 s) on the matched-cardinality sweep $R = 1, \dots, 40$ over the three datasets. Every comparison is read off at **fixed, shared cardinalities** rather than aggregated across R — the earlier draft of this report used medians over the whole sweep for several tables, which blends a method’s cheap- R behaviour with its expensive- R behaviour into one number and can rank two methods in an order neither of their matched- R curves ever shows. That draft is superseded; every table below states the R it was read at.

5.1 Non-negativity holds, exactly, for every cone method

Across all 817 cells and every cone method, the maximum non-negativity violation of the reconstruction is exactly 0. This is structural, not numerical: $c \geq 0$ and $\xi \geq 0$. The POD negative control behaves as [BEE20] §5 predicts and fails: at $R = 8$ it violates on 12.9% of entries on Half-disks and 89.4% on Pellet-Cladding. Driven through the reduced *solve*, POD’s recovered multiplier reaches -195 where every cone method returns exactly 0.

5.2 Precision and cost at matched cardinality

The obvious way to summarize a sweep is a median over R . It is also the wrong way to compare methods: the whole point of a reduced basis is *accuracy for a given number of components*, and a median blends together how a method does at $R = 2$ with how it does at $R = 40$, which is not a comparison anyone asked for and can rank two methods in an order neither of their matched- R curves ever shows. Every number in this section is instead read off at fixed, shared cardinalities — $R = 4$ (the cheap regime) and $R = 16$ (past the initial transient, and still meaningful on all three datasets, see §6) — directly from the same rows the figures in §3–4 are drawn from.

dataset	method	$R = 4$		$R = 16$	
		test err	solves	test err	solves
3D Pellet-Cladding	CPG	0.01737	329	0.00395	1457
	mCPG	0.01737	181	0.00451	631
	ADG	0.05388	94	0.01482	658
	NMF	0.02037	0	0.01426	0
Half-disks of Hertz	CPG	0.07089	217	0.05991	961
	mCPG	0.07043	117	0.03647	375
	ADG	0.07089	62	0.05991	434
	NMF	0.08554	0	0.05692	0
membrane (2-D)	CPG	0.8032	700	0.6312	3100
	mCPG	0.8032	393	0.6286	1479
	ADG	0.7575	200	0.6333	1400
	NMF	0.6649	0	0.6076	0

Bold marks the lowest error and the fewest *constrained* solves per block; NMF’s 0 is never bolded as “cheapest” since it means “uncounted”, not “free” (§3.2). Read this against the median table it replaces: on Pellet-Cladding, CPG and mCPG now visibly *dominate* ADG in error at both cardinalities rather than merely on average, and NMF is not competitive with any of them there at all — the opposite impression a blended number could leave.

The CPG/ADG cost ratio is a function of R alone, not of the dataset. Reporting one “median” ratio hid that it moves — and hid the more interesting fact that, at any fixed R , it is *the same number on all three datasets*:

R	4	8	16	24
solves(CPG)/solves(ADG)	3.50	2.50	2.21	2.14

identically, to three significant figures, on 3D Pellet-Cladding, Half-disks and membrane. Since K_{full} ’s dimension, geometry and conditioning differ enormously across the three (Table in §6), a ratio this exactly reproducible cannot be a property of the data — it is combinatorial, fixed by how many constrained solves each selection rule issues per round (§3.2), and is largest exactly where it matters most: $3.5\times$ at $R = 4$, settling toward $2\times$ by $R \geq 24$.

ADG equals CPG exactly on Half-disks, and the reason is measurable. The table above shows identical errors for CPG and ADG on Half-disks at both cardinalities. This is not approximate agreement: their selected snapshot *sets* are identical at $R = 4, 8, 16, 24$, verified directly against `selected_indices`. The cause is that CPG ranks by raw residual magnitude while ADG ranks by angle on normalized snapshots, and the two rankings coincide whenever snapshot norms are close to uniform — which Half-disks’ are ($\max/\min = 1.09$, $\text{std}/\text{mean} = 2.4\%$) and Pellet-Cladding’s emphatically are not ($\max/\min = 187$, $\text{std}/\text{mean} = 71\%$, where the two methods select different snapshots and land 3–4 $\times$ apart in error). Membrane sits between the two ($1.39\times$) and its selections partially agree.

NMF’s apparent Half-disks advantage is a rank-cliff artefact, not a general edge. At $R = 4$ NMF is worse than CPG/ADG (0.0855 vs. 0.0709); by $R = 16$ it is narrowly better (0.0569 vs. 0.0599). The reason is not that NMF is more accurate — it is that CPG and ADG **freeze**. Their error on Half-disks is exactly 0.0599 at every $R = 16, 17, \dots, 31$ (sixteen bit-identical values), because both are confined to the span of selected snapshots and that span has exhausted the training set’s numerical rank (17 of 31, §6.4) by $R = 16$: no further generator can add anything. NMF is refit from scratch at each R with *optimized* atoms rather than selected snapshots, so it is not confined to that span and keeps improving past the cliff, reaching 0.0075 by $R = 31$ — eight times better than CPG/ADG’s frozen value. On Pellet-Cladding, which never hits its rank ceiling in the sweep, no such reversal occurs and CPG/mCPG beat NMF at every matched R shown.

5.3 mCPG leaves the snapshot cone, and the fraction grows with R

[NDEE22] Remark 4.3’s parenthetical claims ν_r lies “in fact of $\text{Span}^+(\{\theta_{q_n}\})$ ”. It does not hold as stated, and the extent of the failure is itself a function of R that a single summary number would flatten:

dataset	$R = 4$	$R = 8$	$R = 16$	$R = 24$
3D Pellet-Cladding	0.000	0.000	0.250	0.375
Half-disks of Hertz	0.750	0.875	0.938	— [†]
membrane (2-D)	0.750	0.875	0.938	0.958

[†]mCPG self-terminates near $R = 23$ on Half-disks, so no $R = 24$ cell exists (§5.2 note on skips). CPG and ADG are 0.000 at *every* matched R on all three datasets, as they must be: their generators *are* snapshots. What *does* hold for mCPG is the W^+ membership the same sentence claims first, which is the part the construction actually rests on. The extent metric confirms the escape independently and at the same matched cardinalities: at $R = 16$, mCPG’s cone measures $2.80\times$, $22.2\times$ and $0.91\times$ the width of K_{full} on Pellet-Cladding, Half-disks and membrane respectively — exceeding 1 (wider than everything the snapshots generate) on the two datasets where it is also leaving K_{full} most, which is only possible by construction if it has left.

5.4 Conditioning at matched cardinality: no method dominates

Naming one method “best conditioned” from an aggregate would repeat the mistake this section exists to fix. At matched R , away from the numerically-singular region (§6), the ranking is dataset-

specific: ADG leads on Pellet-Cladding at small R but is overtaken by NMF past $R \approx 15$; mCPG is two to four orders of magnitude better conditioned than CPG/ADG on Half-disks, where the latter two are identical to each other; and on membrane ADG holds a modest, consistent edge throughout. The full matched- R table is in §6, which also shows this ranking is not the interesting regularity — the *rate* at which conditioning is lost is set by the dataset, far more than by the method.

5.5 ADG’s pair initialization: mostly irrelevant, and never decisive

Comparing base ADG against `adg_k0` (CPG’s first step) at matched R :

dataset	pair better	$K_0 = \{0\}$ better	identical
3D Pellet-Cladding	0	0	39 of 39
Half-disks of Hertz	0	0	30 of 30
membrane (2-D)	12	27	0 of 39

On two of three datasets the two initializations produce *the same cone at every cardinality*, because the largest-magnitude snapshot already lies in the largest-angle pair. Where they differ, the CPG-style start wins more often. Offline cost is unaffected: the choice moves *which* snapshots are selected, never how many solves it takes. The pair initialization is therefore not earning its departure from the papers on this evidence.

5.6 Agreement between the duplicate transcriptions

CPG’s two independent transcriptions agree on **36 of 36** comparisons. mCPG’s do not: of 18 comparisons, 7 are exactly equivalent, 1 agrees within solver tolerance, 9 produce a *different cone of the same accuracy*, and 1 is genuinely divergent. This is a finding about [NDEE22] Algorithm 2 line 9, whose constrained shift is under-specified — the two implementations resolve it with different solvers, and the resulting cones differ.

6 Conditioning: a closer study

Conditioning deserves its own treatment because it is the quantity that decides whether a cone of a given cardinality is *usable*, independently of how accurate it is. The reduced solve inverts $\hat{A}$ and forms $\hat{B}\hat{A}^{-1}\hat{B}^\top$; recovering the coefficients c from a cone projection is an NNLS problem on Ξ . Both inherit $\kappa(\hat{\Xi}^\top\hat{\Xi})$. A cone that reaches a smaller projection error at a condition number of 10^{17} has not necessarily bought anything.

6.1 What is measured, and on which convention

κ is computed on **column-normalized** generators, and the choice is not cosmetic. [BEE20]’s CPG keeps raw selected snapshots; [NDEE22]’s mCPG normalizes. Measured on raw generators the number would partly report snapshot magnitude spread rather than cone geometry, and the two families would not be comparable. Both are recorded; their ratio shows how much the convention was worth:

membrane_2d — ADG Initialization: largest-angle pair vs $K_0 = \{0\}$
 identical angular-defect rule, batch admission, normalization and stopping; only the first generator differs

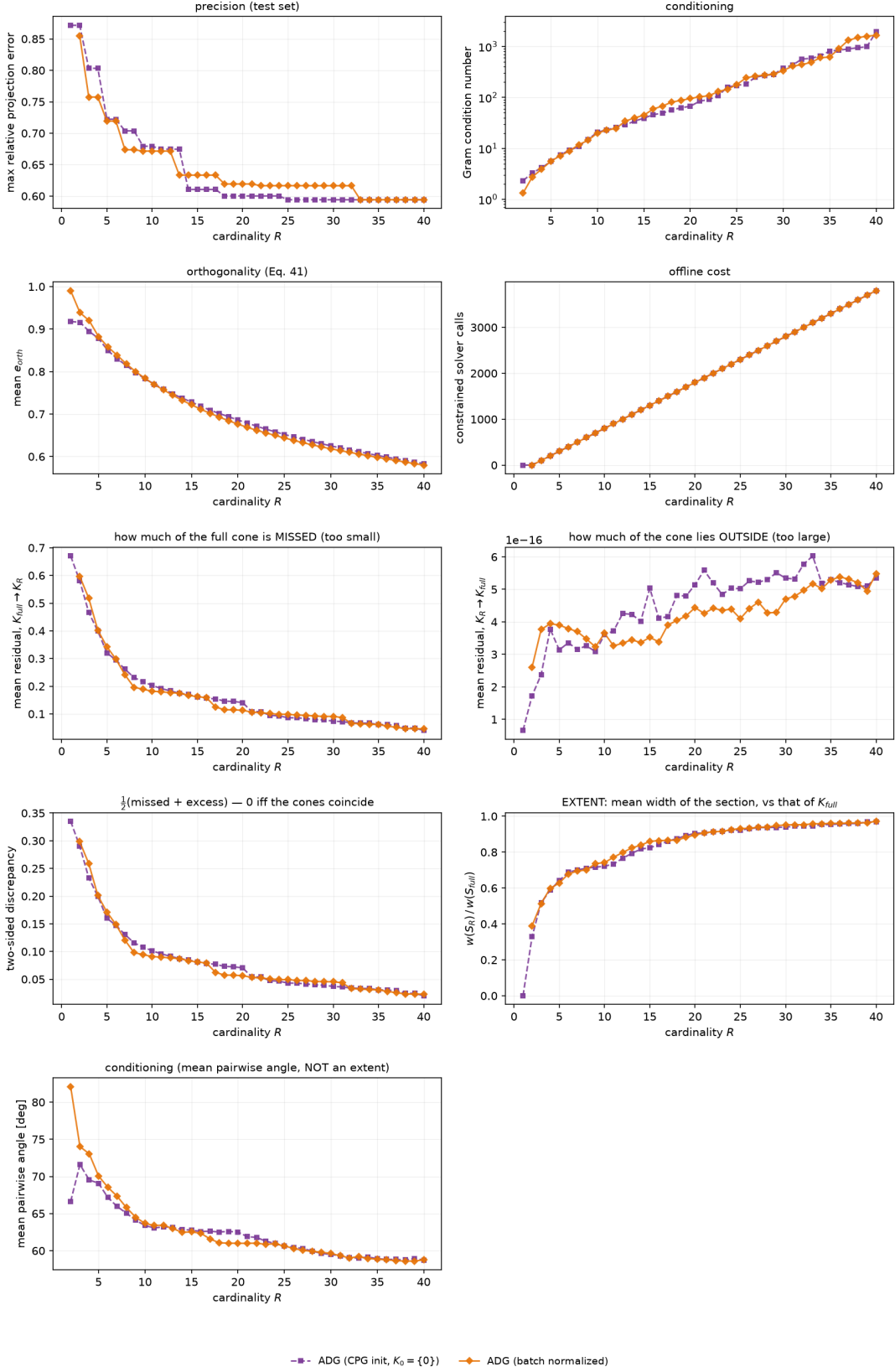


Figure 4: The initialization ablation on membrane — the one dataset of the three where the two arms differ at all. Every panel is the same algorithm apart from the first generator. On the other two datasets the curves coincide exactly at every cardinality.

dataset	method	κ (normalized)	κ (raw)	raw/norm
3D Pellet-Cladding	CPG	9.38×10^6	2.42×10^6	0.26
	mCPG	1.12×10^7	1.12×10^7	1.000
	ADG	8.15×10^4	1.29×10^6	15.8
Half-disks of Hertz	CPG	1.38×10^{17}	5.67×10^{16}	0.41
	mCPG	1.41×10^5	1.41×10^5	1.000
membrane (2-D)	CPG	104	96.8	0.93
	mCPG	98.1	98.1	1.000

mCPG’s ratio is exactly 1.000 on all three datasets, which is the convention confirming itself: its generators are already unit vectors, so normalizing is the identity for it. For ADG on Pellet-Cladding the two differ by $15.8\times$ — had the raw number been reported, ADG’s conditioning advantage there would have been hidden entirely.

6.2 Conditioning at matched cardinality, before any rate is fitted

The rate fitted below is a regression over the whole non-singular range, and while it is a slope rather than a median, it is still a summary that can hide who is ahead *at any one* R . The direct answer to that comes first:

dataset	method	$R = 5$	$R = 10$	$R = 15$	$R = 20$
3D Pellet-Cladding	CPG	9.78×10^3	5.83×10^4	9.44×10^4	8.92×10^6
	mCPG	9.78×10^3	5.83×10^4	8.78×10^4	6.94×10^6
	ADG	2.19×10^2	4.58×10^3	3.21×10^4	7.11×10^4
	NMF	5.44×10^2	4.28×10^4	1.61×10^4	8.73×10^3
Half-disks of Hertz [‡]	CPG	6.92×10^3	2.31×10^6	1.61×10^9	sing.
	mCPG	2.99×10^2	8.10×10^3	5.21×10^6	sing.
	ADG	6.92×10^3	2.31×10^6	1.61×10^9	sing.
	NMF	7.64×10^3	5.24×10^4	7.98×10^5	sing.
membrane (2-D)	CPG	5.99	25.1	52.8	98.1
	mCPG	5.99	25.1	50.0	91.8
	ADG	5.61	20.0	44.6	95.1
	NMF	9.24	14.1	15.7	15.0

[‡] $R = 20$ on Half-disks is past the numerical-rank cliff at $R \approx 17$ (§6.4); the basis is numerically singular there and any ranking would be comparing round-off, so the column is intentionally left blank rather than filled with noise. Bold marks the best-conditioned method in each cell.

No method wins everywhere, and the earlier report’s claim that ADG is simply “the best conditioned of the greedy methods” does not survive contact with this table: it is correct on membrane and at small R on Pellet-Cladding, but **mCPG is two to four orders of magnitude better conditioned than CPG and ADG on Half-disks** (299 against 6923 at $R = 5$; 8.1×10^3 against 2.3×10^6 at $R = 10$), and **NMF overtakes ADG on Pellet-Cladding** once $R \gtrsim 15$. CPG and ADG are identical on Half-disks for the reason given in §5.2 — coincident selection under a near-uniform snapshot spectrum — so the mCPG gap there is not “mCPG beats CPG and ADG” so much as “mCPG picks a different, and much better conditioned, cone.”

6.3 Growth is geometric in R , at a rate set by the dataset

Beyond who is ahead at a given R , the matched- R table above already shows a second, orthogonal question: how fast does each curve *move*. On a log axis the curves are close to straight over most of

Conditioning grows exponentially in R , at a rate set by the dataset

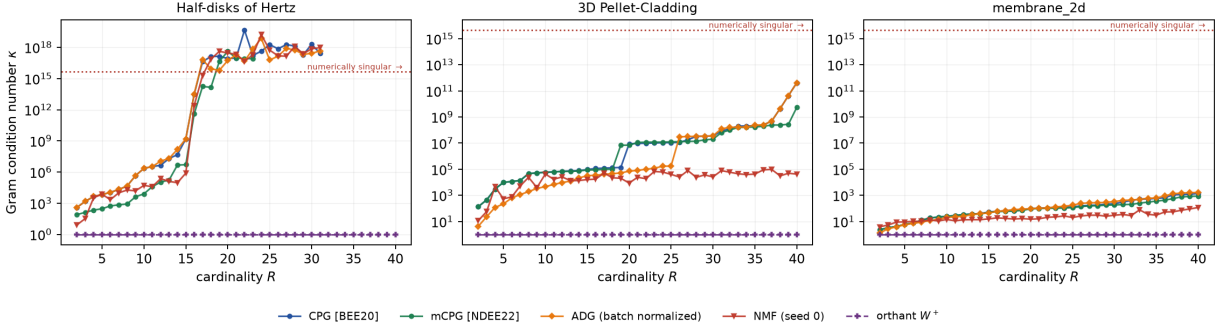


Figure 5: κ against cardinality. Note the log axis — this is the one panel in the benchmark that uses one, because $\kappa \geq 1$ by definition so the zero-dropping failure of a log scale cannot occur, and the quantity spans eleven decades. The orthant sits at exactly 1 at every R : its generators are mutually orthogonal, so it is the best-conditioned admissible cone there is, and it bounds every curve from below.

their range, i.e. κ grows geometrically with each added generator — with one important exception, Half-disks, taken up in §6.4. Fitting $\log_{10} \kappa$ against R over the non-singular range gives the rate in *decades gained per generator*:

dataset	CPG	mCPG	ADG	NMF
3D Pellet-Cladding	0.180	0.157	0.222	0.051
Half-disks of Hertz	0.572	0.705	0.586	0.608
membrane (2-D)	0.063	0.057	0.071	0.027

The rate is a property of the *dataset* far more than of the method: it varies by a factor of ten across datasets and by well under two across methods on any one of them. NMF is consistently the gentlest, which is what one expects from atoms that are optimized rather than selected — it is free to place a generator where it improves conditioning, whereas a greedy must take a snapshot that exists.

6.4 The collapse is the snapshot spectrum, not the algorithm

The Half-disks rates above are misleading taken alone, and the figure shows why: that dataset does not decay geometrically at all. It is flat to $R \approx 15$ and then falls off a cliff, with every method collapsing together within two cardinalities of each other ($R = 17$ for CPG and ADG, 18 for NMF, 19 for mCPG). A single fitted slope averages across that cliff and describes neither side of it.

The explanation is the numerical rank of the snapshot set:

dataset	$\dim \times n_{\text{train}}$	σ_2/σ_1	$\text{rank}_{10^{-8}}$	conditioning behaviour
Half-disks of Hertz	57×31	0.026	17 of 31	flat, then a cliff at $R \approx 16$
3D Pellet-Cladding	7676×47	0.057	41 of 47	smooth staircase, never singular
membrane (2-D)	497×100	0.380	100 of 100	gentle, $\kappa < 10^4$ throughout

Once R exceeds the numerical rank, any further generator is a numerically dependent combination of those already selected, and no selection rule can avoid it: the information is not in the data. This is the single most useful conditioning result in the benchmark, because it is *predictive* — the spectrum is computable before any method runs, and it says where the usable range ends. On Half-disks that is $R \approx 15$, and reporting a Half-disks result at $R = 30$ is reporting round-off.

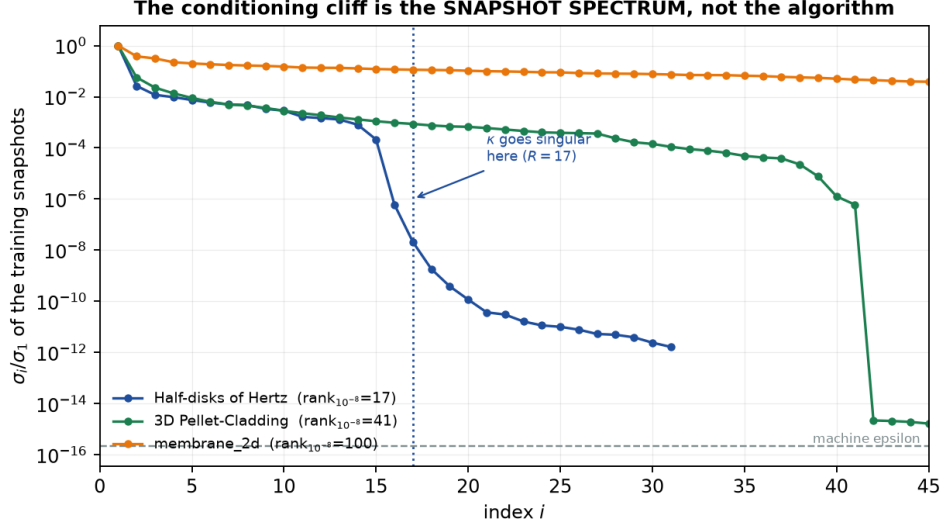


Figure 6: Normalized singular values of the training snapshots. The Half-disks spectrum drops from $\sigma_{14} = 2.1 \times 10^{-4}$ to $\sigma_{15} = 6.0 \times 10^{-7}$ to $\sigma_{16} = 2.1 \times 10^{-8}$ — three decades in two indices. That cliff is at exactly the cardinality where every method’s conditioning explodes.

6.5 Conditioning against ADG’s own stopping criterion

§6.4 located the cliff in R , which requires already knowing the numerical rank in advance. ADG does not need that: at every round it computes $\theta_{\max} = \max_q \text{unresolved } \theta_K(x_q)$, the worst remaining angular defect, and compares $\sin \theta_{\max}$ against its tolerance to decide whether to keep going (§2.3). This is the quantity the algorithm itself watches as it iterates, so the three figures below follow ADG’s own trajectory on each dataset: every marker is one round of `compute_phases()`, placed at the $\theta_{\max}$ ADG reported for that round and at κ read from the matching cardinality in the same matched-cardinality grid as every other figure in this report (so κ is not recomputed, only relocated onto this axis). A handful of markers are labelled with the cardinality R they correspond to, so the two axes used elsewhere — iteration count and angular error — can be read off the same curve.

The practical reading, common to all three: on Half-disks and Pellet-Cladding, $\theta_{\max}$ dropping below roughly one degree is where conditioning risk stops being hypothetical, regardless of what R that happens to correspond to on a given dataset — membrane never gets there within $R \leq 40$, which is itself the finding for that dataset. Because $\theta_{\max}$ is computed anyway, on every round, watching it costs nothing extra and needs no advance knowledge of the snapshot spectrum — unlike the rank criterion in §6.4, which has to be computed once up front and does not update as the greedy proceeds.

6.6 Conditioning and aperture measure the same thing from two sides

The correlation between $\log_{10} \kappa$ and mean pairwise angle is -0.70 , -0.59 and -0.81 on the three datasets: wider cones are better conditioned, consistently and strongly. This is why aperture is retained as a *conditioning* diagnostic in §3.4 and explicitly not as a measure of extent — it tracks κ , which is what [NDEE22] §4 is after when it asks for a well-conditioned Gram matrix, and it does *not* track how much space the cone encloses.

6.7 Which method to prefer, on conditioning grounds

There is no single answer, and the matched- R table in §5.2 and §6 are what replace the single answer an earlier version of this report gave. Three genuinely different regimes hold:

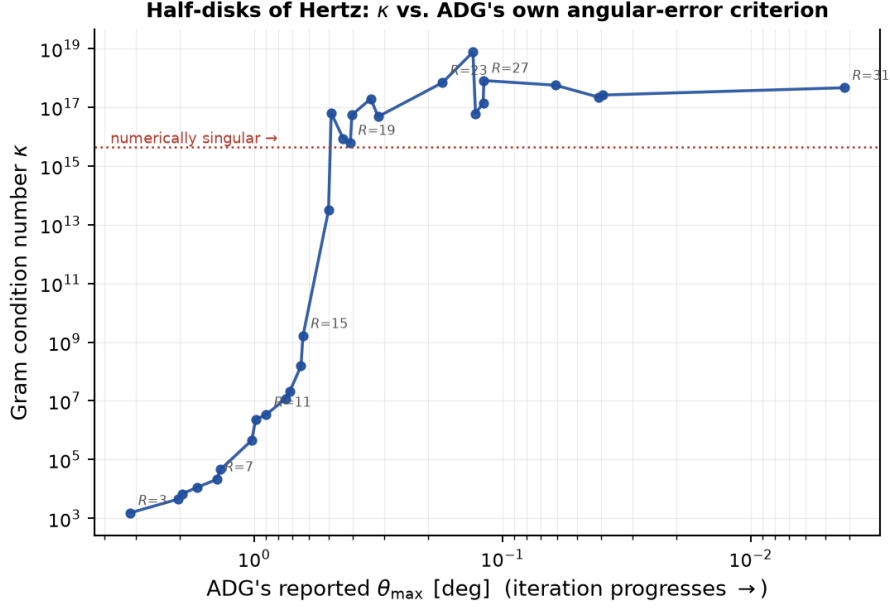


Figure 7: Half-disks of Hertz. κ is flat and modest (10^3 – 10^6) for every $\theta_{\max} \gtrsim 2^\circ$ (through $R = 15$), then rises twelve orders of magnitude while $\theta_{\max}$ moves from about 2° to 0.5° ($R=16$ to 19) — under a factor of four in the criterion itself triggers the entire collapse. Points past the singular line ($R \geq 17$) wobble rather than climb monotonically, confirming they are round-off on an already-singular Gram matrix (§6.4), not a second effect the criterion is tracking.

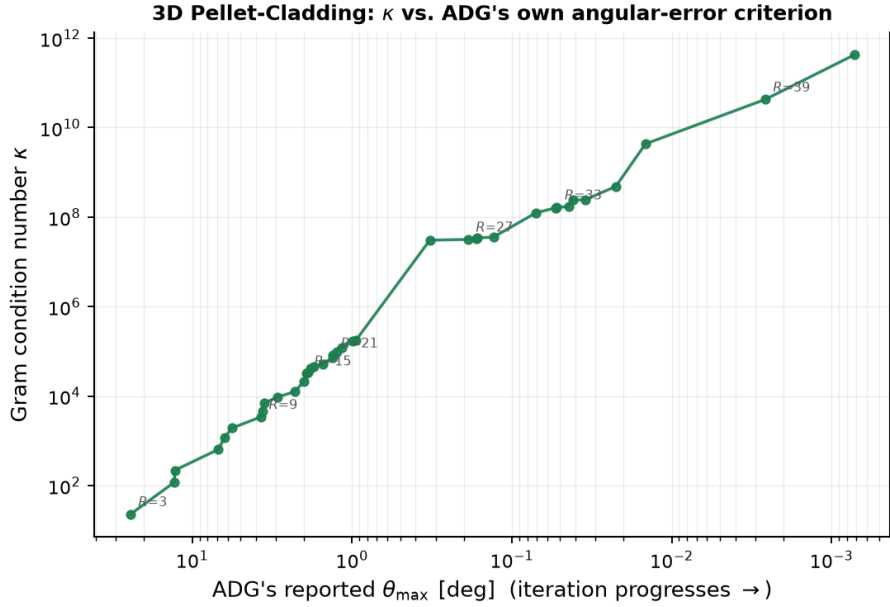


Figure 8: 3D Pellet-Cladding. The curve is smooth and monotonic across nearly five decades of $\theta_{\max}$ (from 24° at $R = 3$ down to 7×10^{-40} at $R = 40$) with no cliff and no crossing of the singular threshold anywhere in the sweep — consistent with its numerical rank (41 of 47) never being exhausted at $R \leq 40$. Every added generator buys a roughly constant fraction of the remaining angular error, which is exactly the geometric decay §6.4's table already reported from the σ_i spectrum, now shown as ADG experiences it round by round.

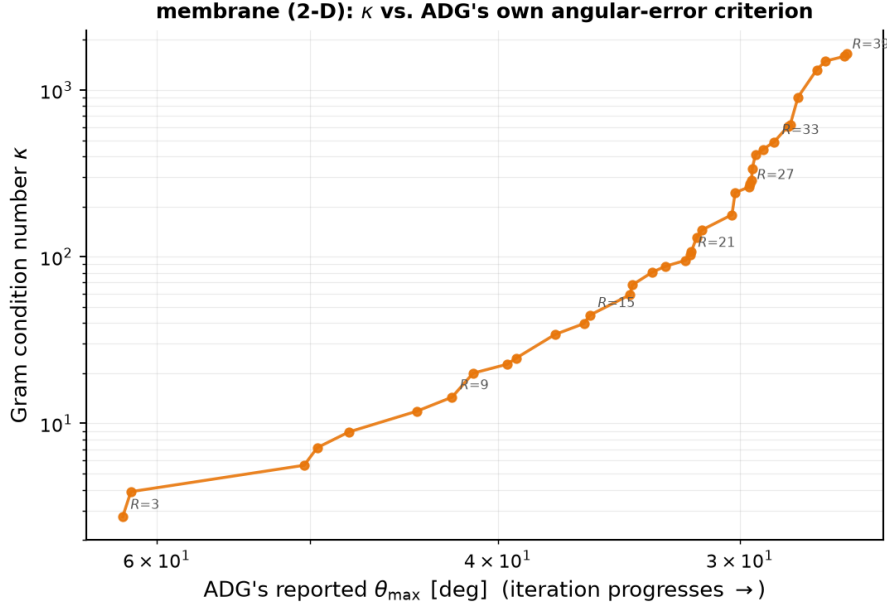


Figure 9: Membrane (2-D). Within $R \leq 40$, $\theta_{\max}$ has fallen only from 62° ($R=3$) to 26° ($R=39$) — axis note: $\theta_{\max}$ still decreases left to right as in the other two panels, but the whole panel occupies the *loose* end of the range those cover. By ADG's own criterion the greedy has barely begun to converge, the same conclusion §5.2 reached from the error curve (membrane is simply not compressible to $R \leq 40$), now visible directly in the algorithm's internal state rather than inferred from its output. κ stays three orders of magnitude below the singular threshold throughout, so this dataset's difficulty is compressibility, not conditioning.

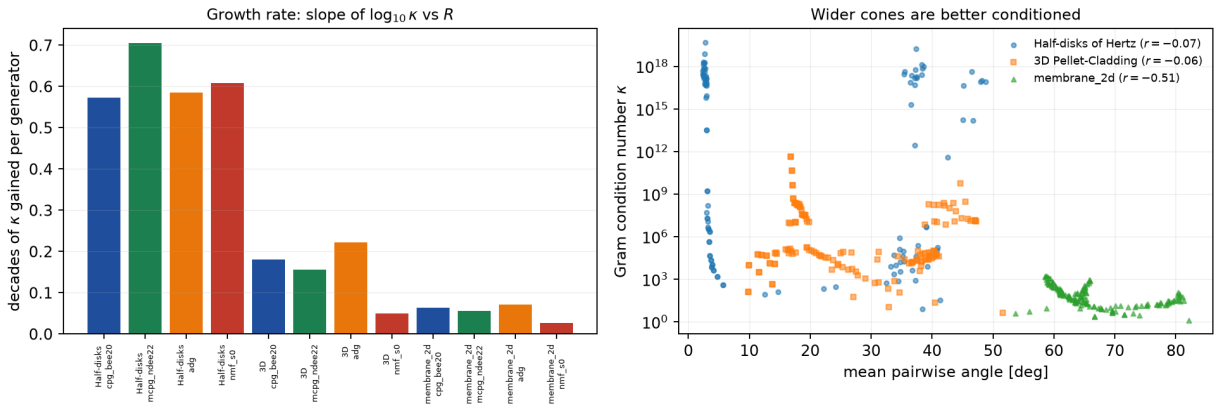


Figure 10: Left: growth rate per method and dataset. Right: κ against mean pairwise angle, pooled over methods and cardinalities.

- **Pellet-Cladding.** ADG is markedly better conditioned than CPG/mCPG at small R — $\kappa = 2.2 \times 10^2$ against 9.8×10^3 at $R = 5$, a factor of 45 — consistent with its selection rule: choosing by *angular* defect rather than residual magnitude is, almost by definition, choosing generators far from the span already held. That advantage over CPG/mCPG persists to $R = 20$ (7.1×10^4 against 8.9×10^6 , a factor of 125), but *NMF* overtakes ADG itself by $R \approx 15$, so “ADG is best” is only true if the comparison set excludes the baseline.
- **Half-disks.** Below the rank cliff, mCPG is dramatically ahead of both CPG and ADG (two to four orders of magnitude, above), which are identical to each other and to no one’s advantage. Past the cliff every method is numerically singular and no ranking is meaningful.
- **Membrane.** ADG holds a modest, consistent edge over CPG and mCPG throughout (10–30%), while NMF is decisively best-conditioned of all four (15–16 against $\kappa \gtrsim 90$ for every cone method by $R = 20$) — unsurprising, since it is not confined to selecting from a snapshot set at all.

The one claim that *does* generalize is the one in §6.4: the numerical rank of the training snapshots bounds every method identically and is computable before any of them runs, which is more actionable than knowing which greedy algorithm has a conditioning edge within that bound.

7 Caveats

1. **No dataset currently drives a reduced solve.** The two 1-D obstacle toys were the only sources shipping A , $B(\mu)$, f and g ; with them removed, the inf-sup family and the solved-error metric report nothing on the three retained datasets. Both remain implemented and generic, and their tests build such a problem themselves.
2. **The orthant’s cost is mCPG’s ordering run,** not the cone assembly, which is free. Tracking mCPG is what buys the comparability; the price is that the cost column measures the tracking.
3. **Extent above 1 is not an error.** It means the cone is wider than K_{full} , which requires leaving it. mCPG and the orthant both do.
4. **The full-cone reference in the solved-error table is not a lower bound.** The reduced solve minimizes over the cone rather than projecting onto it, so a smaller cone can land closer.
5. **Half-disks results are meaningless past $R \approx 15$,** and this is a statement about the data rather than the methods: the snapshot spectrum has numerical rank 17 of 31, so beyond that every generator is a numerically dependent combination of those already held (§6). The usable range is predictable from the spectrum before any method is run.
6. **Membrane errors are large in absolute terms (≈ 0.6)** for every method: 125 snapshots on 497 dual dofs with a moving contact patch is simply not compressible to $R \leq 40$. Comparisons there remain valid; the absolute level is a property of the dataset.
7. **Comparisons are matched-cardinality, not aggregated.** Every ranking in §5–§6 is read at one or a few fixed R from the same sweep the figures use. A median over R was tried first and discarded: it can and did reverse rankings a matched- R reading shows plainly, most visibly for NMF’s apparent Half-disks advantage (§5.2), which turned out to be confined to cardinalities past the training set’s numerical rank rather than a general property of the method.

Reproduction

```
# --datasets defaults to all three; listed here to make the scope explicit.
python -m bench run --datasets fem_lambda physics membrane_2d \
    --subsample 200 --out results
```

```
python -m bench run --datasets fem_lambda physics membrane_2d --deltas \
    --cardinalities $(seq 1 40) --no-infsup --no-determinism \
    --subsample 200 --out results/sweep_dense
python -m bench report --results results --out results/report.txt
python -m bench figures --results results/sweep_dense --out results/figures --separate
python -m bench decrement --cardinality-results results/sweep_dense \
    --tolerance-results results --out results/figures
python -m bench reconstruct --R 8 --separate --out results/figures
```

Main grid 174/210 cells in 919s; dense sweep 817/840 cells in 1848s; 146 tests passing.

A Derivation of the reduced dual problem (Eq. 9)

Step 0 — the HF problem is a constrained minimization

System (1) is the KKT system of

$$\min_u \frac{1}{2} u^\top A u - f^\top u \quad \text{subject to} \quad B u \leq g, \quad (10)$$

with $\lambda \geq 0$ the multiplier of the inequality. A is symmetric positive definite (a stiffness matrix), which is what makes every elimination below legitimate.

Step 1 — substitute the reduced ansatz

Take $u \approx V a$ with $a \in \mathbb{R}^N$ free in sign, and $\lambda \approx \Xi \alpha$ with $\alpha \geq 0$. Testing the equilibrium equation against V and the constraint against Ξ gives the reduced KKT system

$$\hat{A} a + \hat{B}^\top \alpha = \hat{f}, \quad \hat{B} a \leq \hat{g}, \quad \alpha \geq 0, \quad (11)$$

with $\hat{A} = V^\top A V$, $\hat{B} = \Xi^\top B(\mu) V$, $\hat{f} = V^\top f$, $\hat{g} = \Xi^\top g$. Equivalently: minimize $\frac{1}{2} a^\top \hat{A} a - \hat{f}^\top a$ subject to $\hat{B} a \leq \hat{g}$.

Step 2 — eliminate a

This is the step Eq. (9) compresses. The Lagrangian of the reduced problem is

$$L(a, \alpha) = \frac{1}{2} a^\top \hat{A} a - \hat{f}^\top a + \alpha^\top (\hat{B} a - \hat{g}). \quad (12)$$

L is strictly convex in a , so the inner minimum is attained where $\partial_a L = \hat{A} a - \hat{f} + \hat{B}^\top \alpha = 0$, i.e.

$$a^*(\alpha) = \hat{A}^{-1} (\hat{f} - \hat{B}^\top \alpha). \quad (13)$$

Write $s = \hat{f} - \hat{B}^\top \alpha$, so $a^* = \hat{A}^{-1} s$. Substituting,

$$\begin{aligned} L(a^*, \alpha) &= \frac{1}{2} s^\top \hat{A}^{-1} s - \hat{f}^\top \hat{A}^{-1} s + (\hat{B}^\top \alpha)^\top \hat{A}^{-1} s - \alpha^\top \hat{g} \\ &= \frac{1}{2} s^\top \hat{A}^{-1} s - \underbrace{(\hat{f} - \hat{B}^\top \alpha)^\top}_{= s^\top} \hat{A}^{-1} s - \alpha^\top \hat{g} = -\frac{1}{2} s^\top \hat{A}^{-1} s - \alpha^\top \hat{g}. \end{aligned} \quad (14)$$

Expanding $s^\top \hat{A}^{-1} s = \hat{f}^\top \hat{A}^{-1} \hat{f} - 2\alpha^\top \hat{B} \hat{A}^{-1} \hat{f} + \alpha^\top \hat{B} \hat{A}^{-1} \hat{B}^\top \alpha$ and *maximizing* the dual (equivalently minimizing its negative), the constant $\frac{1}{2} \hat{f}^\top \hat{A}^{-1} \hat{f}$ drops and what remains is exactly Eq. (9):

$$\min_{\alpha \geq 0} \frac{1}{2} \alpha^\top Q \alpha - \alpha^\top c, \quad Q = \hat{B} \hat{A}^{-1} \hat{B}^\top, \quad c = \hat{B} \hat{A}^{-1} \hat{f} - \hat{g}. \quad (15)$$

The displacement is then recovered from (13), and $\hat{\lambda} = \Xi \alpha$.

Step 3 — what Q and c mean

Neither is an arbitrary algebraic object.

- $Q = \hat{B}\hat{A}^{-1}\hat{B}^\top$ is the **Schur complement**, i.e. the compliance of the structure *seen from the contact interface*. Entry Q_{ij} is the interface displacement in generator direction i produced by a unit contact force in direction j . It is symmetric positive *semi*-definite.
- $c = \hat{B}\hat{A}^{-1}\hat{f} - \hat{g}$ is the **penetration of the contact-free solution**: $\hat{A}^{-1}\hat{f}$ is the displacement with contact ignored, $\hat{B}$ reads it at the interface, and $\hat{g}$ is the gap. Positive entries are the penetration that contact forces must remove.

So Eq. (15) reads: *choose non-negative contact-force coefficients that undo the penetration c , weighted by the interface compliance Q .*

Step 4 — in what sense it *is* a projection

The objection “this minimizes an energy rather than projecting” is half right. Completing the square in the $\hat{A}^{-1}$ inner product $\langle x, y \rangle_{\hat{A}^{-1}} = x^\top \hat{A}^{-1} y$,

$$\frac{1}{2} \alpha^\top Q \alpha - \alpha^\top c = \frac{1}{2} \|\hat{B}^\top \alpha - \hat{f}\|_{\hat{A}^{-1}}^2 + \alpha^\top \hat{g} - \frac{1}{2} \|\hat{f}\|_{\hat{A}^{-1}}^2, \quad (16)$$

verified numerically to 2.8×10^{-14} over random $\alpha \geq 0$. So the reduced solve *is* a projection — of the load $\hat{f}$ onto the cone of attainable reactions $\{\hat{B}^\top \alpha : \alpha \geq 0\}$, in the **energy (compliance) metric** $\hat{A}^{-1}$, shifted by the linear gap term $\alpha^\top \hat{g}$. When $\hat{g} = 0$ it is exactly that projection.

This is precisely why solved error and projection error are different functionals of the same cone, as §3.6 claims. The metric Π_K used everywhere else in this benchmark is the *Euclidean* projection of the *true multiplier* onto W_R^+ . The solve performs an $\hat{A}^{-1}$ -weighted projection of the *load*, offset by the gap. Two different objects, two different inner products: a cone can be good for one and poor for the other.

Step 5 — a caution about solving (15)

Q is a Schur complement, so its rank is limited by the interface rather than by R : with more cone generators than the contact set can distinguish it is *singular*. On the synthetic obstacle problem used to test this metric, Q is rank-deficient in 60% of cells, one at rank 4 of 6 with $\kappa(Q) = 3.7 \times 10^{16}$.

That defeats a naive bound-constrained solver. L-BFGS-B reports **status=0** (converged) after eight iterations on such a cell while its KKT residual is still 4×10^{-3} , landing on a worse objective than the true optimum in 10% of cells. Because it reports success, nothing downstream notices. Writing (15) as a bounded least-squares problem via the eigen-decomposition $Q = W\Lambda W^\top$ — where NNLS is an exact active-set method — reaches a KKT residual of 1×10^{-14} on the same cell. The implementation runs both and returns the lower objective, so it can never be worse than the reference it mirrors.